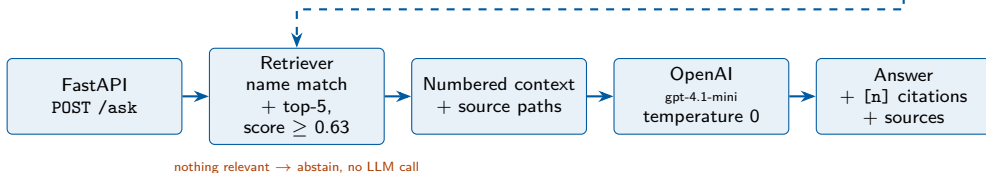


1 | Architecture & data flow

Indexing (once: scripts/build_index.py)



Question answering (per request)



- Python 3.12, FastAPI, Pydantic, sentence-transformers, Chroma, OpenAI SDK
- Plain Python pipeline (no LangChain): each step is one small module
- Endpoints: /ask, /retrieve (no LLM), /health, Swagger /docs
- Docker: model and index built into the image, key only at runtime

2 | Embedding & retrieval strategy

Documents: one entity = one unit of meaning

- Scope: *banking accelerator* (24 entities) + 3 common entities; parents merged (Account: 4 files, 120 attributes)
- Chunks follow entity boundaries, not fixed token windows:
 - overview: description, inheritance, outgoing relationships
 - referenced_by: incoming relationships
 - attributes: 12 per chunk, each starts with Entity: <name>
- Split because whole entities reached 700 tokens (limit 512); largest chunk now 416

Embedding model and vector store

- BAAI/bge-small-en-v1.5: local, CPU, 384 dim., 512-token input
- Chroma: vectors, text and metadata together; cosine; persistent on disk

Retrieval per question

1. **Entity-name match** (“financial product” = FinancialProduct):
always add overview + referenced_by of up to 3 named entities
2. **Vector search**: top- k = 5 (+ BGE query instruction)
3. **Threshold**: keep cosine score ≥ 0.63
4. Merge without duplicates; empty $\rightarrow$ abstain

Why these choices (measured)

- “How does a Contact relate to an Account?”: vector search alone returned only attribute chunks $\rightarrow$ name matching adds the relationship chunks
- Top-1 scores: in-scope ≥ 0.66 , off-topic $\leq 0.61 \rightarrow$ threshold 0.63
- Evaluation script: 17/18 checks.
Miss: “assets pledged to secure a loan” does not find Collateral

3 | Relationships, reliability & testing

Relationships

- Source: explicit relationships list in the official manifests
- Stored per entity: outgoing + incoming
- Written as text:
`Contact.parentCustomerId`
→ `Account.accountId`
- Audit links (`createdBy`, `ownerId`) summarized separately
- Named entities: both sides retrieved
- No graph DB: 27 entities, one-hop questions

Reducing hallucinations

- No relevant chunk → fixed answer, **no LLM call**
- Prompt: only the `<context>`; no invented entities, attributes or links; fixed sentence if context is insufficient; say when lists are partial
- Temperature 0, numbered blocks, `[n]` citations, source file paths
- Checked: all 28 attributes and 5 relationships in demo answers exist in the data
- Example: no direct `Contact` → `Organization` link → answer says so

Testing

- 58 pytest tests, ~6s, no network, no OpenAI
- Fakes: keyword embedder, in-memory Chroma, fake OpenAI client
- Parser, chunks, retrieval (name match, threshold, off-topic), RAG service, API (422/503)
- Retrieval evaluation on the real index

Limitations

- Small scope; no reranker or keyword search
- Attribute lists can be partial
- Threshold from a small sample
- Depends on OpenAI; no auth